

Digital System Design

Course Project — Phase 1: Proposal

Design and Simulation of an 8-bit Arithmetic Logic Unit (ALU)

Course	Digital System Design (DSD)
Phase	Phase 1 — Topic Selection & Proposal
Project Topic	8-bit Arithmetic Logic Unit (ALU)
Submission	Week 9 / Week 10

1. Problem Statement and Motivation

Modern computing systems depend entirely on the Arithmetic Logic Unit (ALU) as the core computational engine of any processor. Every mathematical calculation and logical decision made by a digital system is executed by an ALU. Despite its central importance, the internal design of an ALU remains abstract to many students of digital systems.

This project aims to bridge that gap by designing, implementing, and simulating a functional 8-bit ALU from the ground up using Hardware Description Language (HDL). The ALU will accept two 8-bit binary operands and a 3-bit operation-select signal, execute one of eight supported operations, and output an 8-bit result along with four status flags.

Designing an ALU from scratch provides direct, hands-on experience with combinational logic design, modular HDL coding, simulation-based verification, and FPGA implementation — all core competencies of the Digital System Design course.

2. Proposed Approach and Design Methodology

2.1 System Overview

The ALU will accept two 8-bit operands (A and B) and a 3-bit operation-select signal (OpSel). Based on OpSel, the ALU executes the corresponding operation and drives an 8-bit result (Y) along with four status flags (Zero, Carry, Overflow, Negative).

2.2 Supported Operations

The ALU will support the following eight operations, selected via a 3-bit OpSel signal:

OpSel (3-bit)	Operation	Description
000	ADD	$A + B$ (Addition)
001	SUB	$A - B$ (Subtraction)
010	AND	$A \text{ AND } B$ (Bitwise AND)
011	OR	$A \text{ OR } B$ (Bitwise OR)
100	XOR	$A \text{ XOR } B$ (Bitwise XOR)
101	NOT	NOT A (Bitwise complement of A)
110	SHL	$A \ll 1$ (Logical Shift Left)
111	SHR	$A \gg 1$ (Logical Shift Right)

2.3 Status / Flag Outputs

Along with the 8-bit result, the ALU will generate four status flags:

- Zero Flag (Z) — asserted when the result equals 0x00.
- Carry Flag (C) — asserted when an arithmetic operation produces a carry-out from the MSB.
- Overflow Flag (V) — asserted when a signed arithmetic overflow is detected.
- Negative Flag (N) — asserted when the MSB of the result is 1 (two's complement negative).

2.4 Design Methodology

The project follows a top-down, modular design approach:

- Specification: Define all inputs, outputs, and functional behaviour precisely before writing any HDL code.
- Modular HDL Design: Implement separate sub-modules for the Arithmetic Unit, Logic Unit, Shifter Unit, and a top-level multiplexer that selects the correct output.
- Unit Simulation: Verify each sub-module independently using individual testbenches before integration.
- System Integration: Combine all modules into the top-level ALU entity and run a complete system-level simulation.
- FPGA Synthesis: Synthesise and implement the design on the lab FPGA board to verify real hardware behaviour (subject to equipment availability).

3. Tools and Technologies

Category	Tool / Technology
Hardware Description Language	VHDL / Verilog (as directed by instructor)
Simulation Environment	ModelSim / Questa or Xilinx Vivado Simulator
Synthesis & FPGA Tool	Xilinx Vivado or Intel Quartus Prime
Target Hardware	FPGA Development Board (lab-provided)
Documentation	TeXstudio (LaTeX) for final report; MS Word for interim documents
Version Control	Local backups; Git (optional but recommended)

4. Preliminary Timeline

Week(s)	Phase	Key Tasks
9 – 10	Phase 1 & 2	Topic selection, proposal writing & submission, literature review, background study of ALU architectures and HDL
11 – 12	Phase 3	System specification, block diagrams (high & low level), state/truth tables, design document finalisation
13 – 14	Phase 4 & 5	HDL coding of all sub-modules, unit testbenches, system-level simulation, debugging, and test report
15	Phase 6 & 7	Final report writing (LaTeX), slide preparation, rehearsal, and live demonstration to instructor

5. Division of Responsibilities

The table below outlines the planned distribution of work. All members are expected to contribute equally to documentation, testing, and the final presentation.

Team Member	Primary Responsibility
Muhammad Sharjeel	Literature review, system specification, and design document preparation (Phases 2 & 3)
Dawood Asif	HDL coding of Arithmetic Unit and Logic Unit sub-modules; unit testbenches (Phase 4)

Abdulrehman	HDL coding of Shifter Unit and top-level ALU integration; system testbench and test report (Phases 4 & 5)
Abdullah Masood	Final report writing, slide preparation, rehearsal, and live demonstration (Phases 6 & 7)

Note: Responsibilities are preliminary and may be adjusted based on individual strengths and project requirements as they evolve.

6. Conclusion

This proposal describes the design and simulation of an 8-bit Arithmetic Logic Unit as the course project for Digital System Design. The ALU will support eight fundamental operations controlled by a 3-bit select signal, and will produce an 8-bit result together with four status flags. The entire design will be implemented in HDL, verified through simulation testbenches, and demonstrated on an FPGA board where possible.

The project provides an appropriate level of technical challenge while remaining clearly scoped and achievable within the given timeline. We look forward to the instructor's feedback and request approval to proceed to Phase 2.

Submitted by:

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Instructor Approval: _____ *Date: 2 May 2026*